

Foundation (Jalapeno)

- Q1.** What is the greatest common factor of $6x$ and $9x$?
(A) $3x$
(B) 6
(C) 9
(D) x
(E) 1
- Q2.** Factor out the greatest common factor from $4x + 8y$
(A) $4(x + 2y)$
(B) $2(2x + 4y)$
(C) $8(x + y)$
(D) $2(2x + 4y)$
(E) $4(x + 2y)$
- Q3.** What is the greatest common factor of 12 and 15?
(A) 3
(B) 5
(C) 1
(D) 12
(E) 15
- Q4.** Simplify the expression $9x^2 + 18x$ by factoring out the greatest common factor
(A) $9x(x + 2)$
(B) $18x(x + 1)$
(C) $3x(3x + 6)$
(D) $x(9x + 18)$
(E) $3(3x^2 + 6x)$
- Q5.** Factor out the greatest common factor from $20x^2 + 16x$
(A) $4x(5x + 4)$
(B) $4(5x^2 + 4x)$
(C) $20x(x + 4)$
(D) $16x(x + 5)$
(E) $8(2.5x^2 + 2x)$
- Q6.** What is the greatest common factor of 24 and 30?
(A) 6
(B) 12
(C) 2
(D) 3
(E) 1
- Q7.** Simplify the expression $15x^3 + 25x^2$ by factoring out the greatest common factor
(A) $5x^2(3x + 5)$
(B) $5x(3x^2 + 5x)$
(C) $25x(x^2 + 1)$
(D) $15x^2(x + 5)$
(E) $5(3x^3 + 5x^2)$
- Q8.** Factor out the greatest common factor from $8x + 12y$
(A) $4(2x + 3y)$
(B) $2(4x + 6y)$
(C) $8(x + y)$
(D) $12(x + y)$
(E) $4(2x + 3y)$

Open Ended Questions (FRQ)

- Q9.** Simplify the expression $12x^4 + 16x^3$ by factoring out the greatest common factor

- Q10.** Factor out the greatest common factor from $9x^2 + 18x + 27$

Chef's Tips

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- Tip 1: Ensure students understand the concept of greatest common factor
- Tip 2: Remind students to factor out the greatest common factor from all terms
- Tip 3: Encourage students to simplify expressions after factoring